

Wandering Black Hole Feeds on Its Gravitational Wake

A 35,000-solar-mass black hole accreting from the dense wake it creates while moving off-nuclear.

arXiv:2608.10719; multiwavelength evidence for off-nuclear intermediate-mass black hole accretion.

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[arXiv:2608.10719](#) [doi:10.48550/arXiv.2608.10719](#)



Background

Intermediate-mass black holes may seed supermassive black holes, but many are expected to wander far from galactic nuclei. Their fuel supply has remained uncertain because normal fueling channels are largely inaccessible off-nuclear.

Direct observational evidence for gravitational-wake accretion onto wandering black holes has been elusive.

What this paper is selling

This paper reports a 35,000-solar-mass wandering black hole accreting through a gravitational wake. Multi-epoch spectroscopy reveals upstream flow, downstream wake, and optically thick absorbers driving changing-look broad-line variability.

Key results to remember

- Broad lines, compact continuum, variability, and power-law SED confirm black hole nature.
- Three gas components: blueshifted upstream flow, redshifted dense wake, and thick absorbers.
- Absorbers inside capture radius cause rapidly changing-look broad-line emission.

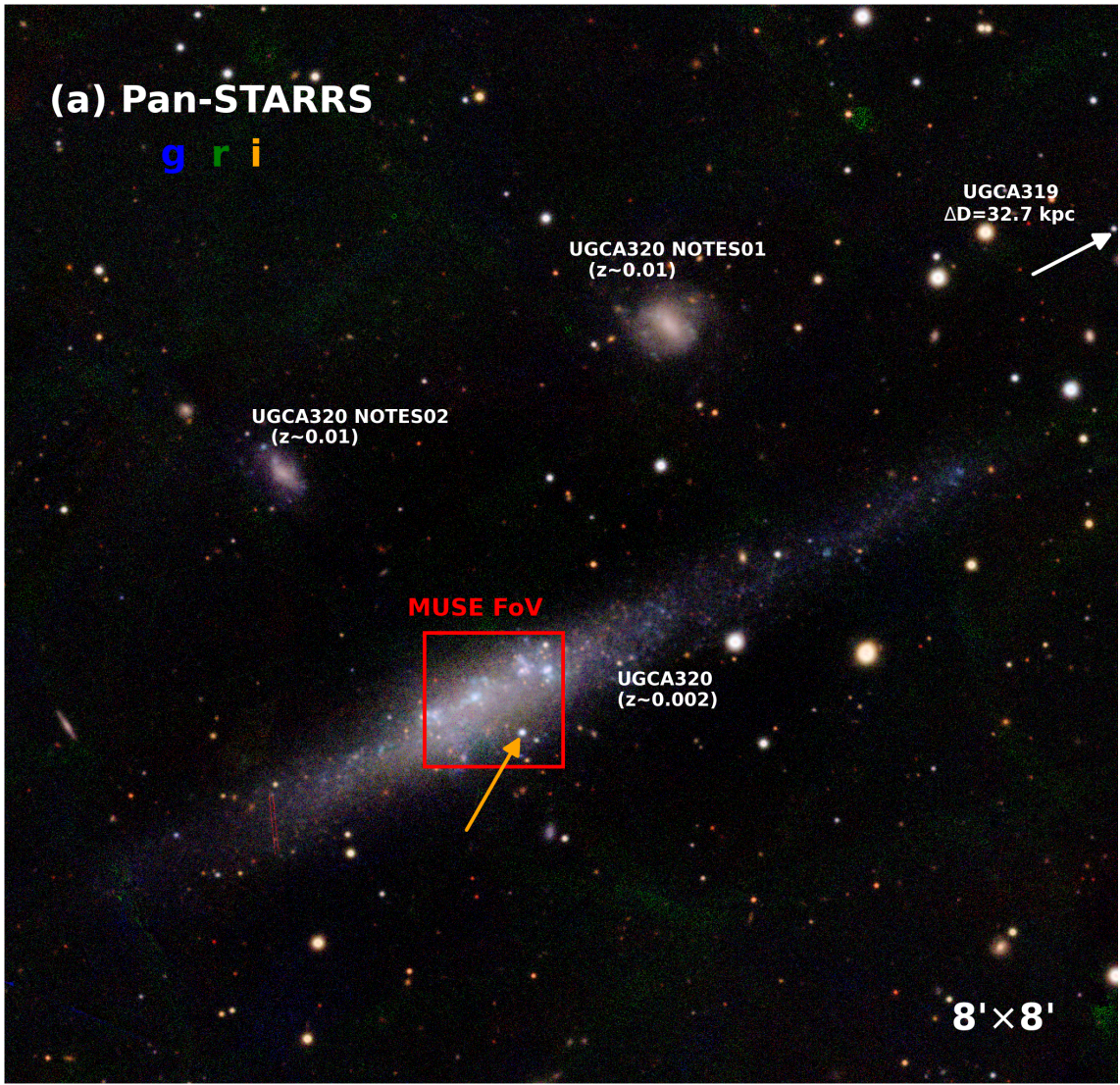


Fig. 1. HST and MUSE views pinpoint the off-nuclear IMBH and reveal its velocity field within UGCA 320.

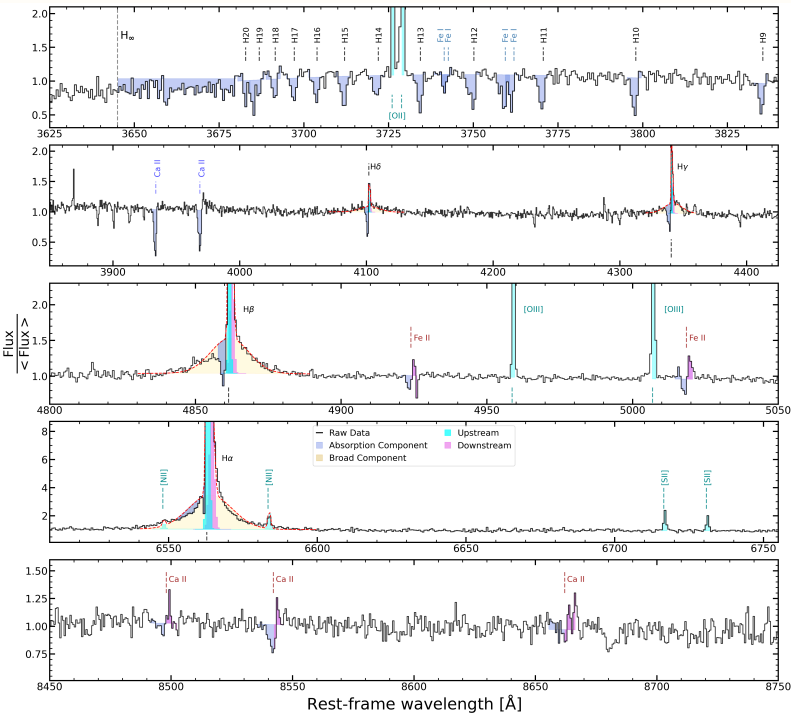


Fig. 3. X-Shooter spectra decompose blueshifted inflow, redshifted wake, and broad AGN emission components.